Program: **Diploma in Automation and Robotics**	
Course Code :RAPC 201	Course Title: **Mechanics of Materials**
Semester :**3**	Credits: **3**
Course Category: **Program Core**	
Periods per week: **3 (L:3, T:0, P:0)**	Periods per semester: **45**

Course Objectives:

- To Understand the concept of stress and strain and their relationship
- To study the concept of shearing force and bending moment due to external loads acing on beams and to compute deflection and slope on it
- To determine stress and deformation in circular shafts and helical spring due to torsion
- To study the concept buckling in determining the strength of columns
- To study the concept of strain energy for different types of loadings
- To study the selection of materials used for making components of Robot

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basic principles and theorems of Engineering Physics		Physics	Secondary school
Basics of differentiation, integration and trigonometry		Mathematics	Secondary school
Concept of equilibrium of forces and concept of moment of inertia	1003	Applied Physics-I	I
Basic concept of engineering materials	1004	Applied chemistry	I

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Apply the concepts of stress and strain in bars, cylindrical and spherical shells	10	Applying

CO2	Applythe concept of bending moment, shear force and deflection in beams	13	Applying
CO3	Apply basic equations of simple torsion in designing of shafts and helical springs	12	Applying
CO4	Selection of materials and its suitability as per Robotic applications.	8	Understanding
	Series Test	2	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3	2					
CO3	3	2					
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Understand the concepts of stress and strain in bars, cylindrical and spherical shell**		
M1.01	Understand basic mechanical properties of materials	1	Understanding
M1.02	Understand the concept of stress and strain	2	Understanding
M1.03	Discuss the stress and strain in composite section	2	Applying
M1.04	Discuss stress and strain relationship	3	Applying
M1.05	Apply hoop stress and longitudinal stress in cylindrical and spherical shells	2	Applying
Simple stress and strain: Concept of rigid body, deformable body, mechanical properties of materials -rigidity, elasticity, plasticity, compressibility, hardness, toughness, stiffness, ductileness, brittleness, malleability, creep and fatigue – definition of stress and strain- tensile, compressive, shear, crushing, thermal stresses, & corresponding strains – stress - strain relationship - elastic limit and limit of proportionality, Hook’s Law –Problems on Direct Stress & Linear Strain- Stress- Strain curve for Ductile material and Brittle material- factor of Safety. stresses in varying section -stresses in composite section - simple problems. Elastic Constants - Lateral Strain, Poisson’s ratio, Bulk Modulus, Shear Modulus, Volumetric Strain - Relation between elastic constants- Problems on elastic			

constants. Hoop stress-Longitudinal Stress in thin cylindrical & spherical shells subjected to internal pressure. -Problems on thin cylindrical shells.			
CO2	**Apply the concept of bending moment, shear force and deflection in beams**		
M2.01	Understand loadings on various beams	3	Understanding
M2.02	Asses shear force and bending moment diagram of simple loading	4	Applying
M2.03	Understand the concept of simple bending	2	Understanding
M2.04	Discuss the deflection and slopes of beams	4	Applying
	Series Test – I	1	
Beams - Types of beams - cantilever beam- simply supported beam- over hanging beam-fixed beam and continuous beam - types of loading - concentrated or point load- uniformly distributed load and uniformly varying load – support reaction calculation. **Shear force and bending moment diagrams:**Shear force and bending moment diagrams of cantilever beams - point load, uniformly distributed load and combination of point load and uniformly distributed load - simply supported beam - point load- uniformly distributed load and combination of point load and uniformly distributed load - maximum bending moment on the section -point of contraflexure. **Bending and deflection of beams**:Theory of simple bending – Assumptions – Relationship between bending moment, bending stress and radius of curvature- simple problems – Deflection of beams – slopes and deflection of simply supported beam and cantilever beams subjected to symmetrical point load and uniformly distributed load (No derivations).			
CO3	**Apply basic equations of simple torsion in designing of shafts and helical springs**		
M3.01	Understand the concept of simple torsion	1	Understanding
M3.02	Apply torsion equations to solid and hollow shafts	2	Applying
M3.03	Study different types of springs and design terms related to springs	2	Understanding
M3.04	Discuss stress induced, stiffness and deflection of helical spring	3	Understanding
M3.05	Explain buckling theory of columns	4	Applying
Torsion: Introduction to pure torsion – assumption in theory of pure torsion -Torsion equation - Angle of Twist, Polar Moment of Inertia - Power Transmitted by a shaft, axle of solid and hollow sections subjected to Torsion - Comparison between Solid and Hollow Shafts subjected to pure torsion- simple problems to calculate strength and power transmitted. **Spring**: Introduction - types of spring - leaf spring - helical springs - closely coiled and open coiled helical spring with round wire – spring index - formulae for deflection-stiffness- torque and energy stored (noderivation) -simple problems on closely coiled helical springs subjected to an axial load to find out diameter, stress induced, stiffness and deflection. **Columns:** Concept of column, modes of failure - Types of columns - Buckling load, crushing load - Slenderness ratio - Factors effecting strength of a column - End restraints -			

Effective length - Strength of column by Euler and Rankine formula (without derivation) – Simple problems to calculate buckling load.			
CO4	**Selection of materials and its suitability as per Robotic applications.**		
M4.01	Discuss the concept of strain energy under different loading conditions	4	Understanding
M4.02	Discuss the building materials used for robotics	4	Understanding
	Series Test – II	1	
Strain Energy-Types of loading-Sudden, Gradual and Impact Load-resilience, proof resilience and modulus of resilience-Equation for strain energy stored in a body when the load is gradually applied and suddenly applied – problems. **Robotics /Design Basics/Building Materials** – Metal, synthetic materials and composite materials – properties of steel from steel construction – properties of wrought aluminum alloys – Mechanical Properties of stainless steel – Designation of Titanium alloys – structural examination of titanium alloys – case hardening of titanium and alloys- How to build robots using advanced materials – Case study.			

Text / Reference

T/R	Book Title/Author
T1	Bansal, R.K., "Strength of Materials", Laxmi Publications (P) Ltd., 2016
T2	SS Bhavikatti ,Strength of materials –Vikas Publishing House
R1	Ramamurtham. S., “Strength of Materials”, 14th Edition, Dhanpat Rai Publications, 2011
R2	Mikell Grover ., — Industrial Robotics , McGraw Hill , 2016.
R3	S.S Rattan, “Strength of Materials”, McGraw Hill, 2nd edition, 2011.